slam-checkpoint-4

Frame transformation:

$\begin{pmatrix} x_v \\ y_v \\ \psi_v \end{pmatrix}$: coordinates of car wrt to global frame which are origin for car's local frame

$(x_{g,j}, y_{g,j})$

↳ coordinates of j^{th} landmark (cones) wrt global frame

$$\begin{pmatrix} x_{j,l} \\ y_{j,l} \end{pmatrix} = \begin{pmatrix} r_j \cos\theta_j \\ r_j \sin\theta_j \end{pmatrix}$$

coordinates of jth cone in car's frame

$$\begin{pmatrix} r_j \\ \theta_j \end{pmatrix} = \begin{pmatrix} \sqrt{x_{j,l}^2 + y_{j,l}^2} \\ \operatorname{atan2}(y_{j,l}, x_{j,l}) \end{pmatrix}$$

$$\begin{pmatrix} r_j \\ \theta_j \end{pmatrix} = \begin{pmatrix} \sqrt{(x_{j,g}-x_v)^2 + (y_{j,g}-y_v)^2} \\ \operatorname{atan2}(y_{j,g}-y_v, x_{j,g}-x_v) - \psi_v \end{pmatrix}$$

covariance matrix for motion update

Motion model:

$$x_t = F x_{t-1} + B u_t + N(0, Q_t)$$

for our check point we are not considering noise

$$\begin{pmatrix} x_t \\ y_t \\ \psi_t \end{pmatrix} = \begin{pmatrix} x_{t-1} \\ y_{t-1} \\ \psi_{t-1} \end{pmatrix} + \begin{pmatrix} \cos(\psi) & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_x \\ u_y \\ \dot{\psi} \end{pmatrix} \Delta t$$

non-linear model

↳ from wheels (rpm) & $\dot{\psi}$ from IMU

↳ yawrate

↓ linearise

$$G_t = \begin{pmatrix} 1 & 0 & (u_x \sin(\psi) - u_y \cos(\psi))\Delta t \\ 0 & 1 & (u_x \cos(\psi) - u_y \sin(\psi))\Delta t \\ 0 & 0 & \Delta t \end{pmatrix}$$

↳ Jacobian

motion update
↓
from x_{t-1}, u_t get x_t
↓
linearise the model

Measurement Update:

$$z_t = H x_t + V_t$$

observations (sensors)

$$P_{t|t-1} = F_t P_{t-1|t-1} F_t^T + Q_t$$

error in state

taking measurements into account for precision

so our model becomes

$$\hat{x}_{t|t} = x_{t|t-1} + K_t (z_t - H_t x_t)$$

$H_t \equiv G_t$

Jacobian

from hear we will get the pose of the car $\Rightarrow$ localisation done

$$K_t = (P_{t|t-1} H_t^T) (H_t P_{t|t-1} H_t^T + R_t)^{-1}$$

covariance matrix for measurement update

z_t: observations (sensor) in car's frame

But for our check point we are having observations (cones) are in ground frame $\Rightarrow$ conversion
(ground_truth/track/viz)

$H_t \hat{x}_t$: the actual pose of the cones now transformed into car's frame using global coordinates

the predicted pose using transformations

Data Association

making the obs by perseption (z_t: in car's frame) of cones accurate wrt $H_t \hat{x}_t$ (actual coordinates of cones wrt car's frame)